Abstract

Abstract I

We specify a reproducible advanced literature-review workflow for rapidly changing domains in which a single query does not adequately represent methodological or temporal variation. The exemplar is configured for exoplanet atmospheric composition and separates foundation, James Webb Space Telescope, and molecular-detection phases.

The design is informed by systematic-review reporting practice [@page2021prisma], but this repository release is a methods exemplar rather than a completed systematic review: its default corpus is synthetic and its phase boundaries, search coverage, and domain interpretation require review for any live application.

Abstract II

The pipeline combines multi-engine retrieval, canonical de-duplication, deterministic screening, optional LLM-assisted classification, full-text assessment, bibliometrics, knowledge-graph construction, visualization, reproducibility assessment, and export. Each stage has a declared input, output, method reference, and definition of done in `methods_pipeline.yaml`. The numbered scripts are thin adapters; reusable computation is tested in `src/`.

The manuscript is generated from configuration and recorded artifacts. Offline runs use a clearly labelled synthetic fixture corpus with reserved identifiers and generated authors; they exercise pipeline behavior and are not evidence about {{SEARCH_TERM}}. Live runs must retain retrieval reports, source provenance, claim classifications, and the exact configuration needed to interpret reported results.

Abstract III

The contribution is therefore infrastructural: it makes phase-aware search design, evidence boundaries, accessibility metadata, and reproducibility checks explicit while leaving domain claims subject to source-backed review.

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